

Introduction to Sport and Exercise Science

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Introduction to Sport and Exercise Science (A10744)

Short Title:	ISES	
Faculty	Science and Computing	
Department	Sport and Exercise Science	
Cluster	Sports Technology	
Author	ASTOREY	
Credits	5	Level: Introductory

Description of Module / Aims

The aim of this module is to develop students' knowledge and understanding of determinants of sports and exercise performance from the perspectives of the core human movement sciences: physiology, biomechanics and psychology.

Programmes

	BA (Hons) in Exercise and Health Studies (WD_HEXHS_B)	1 / 1 / M
SPOR-0044	BSc (Hons) in Software Systems Development (WD_KDEVP_B)	2 / 3 / E
	BSc (Hons) in Sports Coaching and Performance (WD_HSPOR_B)	1 / 1 / M
SPOR-0044	BSc in Software Systems Development (WD_KCOMC_D)	2 / 3 / E
	Bachelor of Business in Recreation and Sport Management (WD_HRECL_D)	1 / 1 / M

Indicative Content

- Anthropometry: stature, body weight, body composition, somatotypes
- Biomechanics: describing motion, force, acceleration, velocity, momentum, torque, levers, work, power.
- Motor behaviour: closed skills, open skills, skill acquisition
- Physiology: components of fitness, aerobic capacity, movement efficiency, aerobic and anaerobic thresholds, strength, muscular endurance, flexibility, anaerobic capacity
- Psychology: personality, anxiety, stress, arousal, motivation, self efficacy

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the anthropometrical, biomechanical, physiological and psychological attributes necessary for success in a range of sports.
2. Define the key anthropometrical, biomechanical, physiological and psychological terms associated with performance ability in sport and exercise.
3. Search for and retrieve appropriate information from trustworthy sports and exercise science sources.

Learning and Teaching Methods

- Lectures
- Tutorials (with problem/enquiry-based learning and group presentations)
- Computer-based tutorials

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Assignment</i>	50%	1,2,3
<i>Presentation</i>	50%	1,2

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Tutorial	24	
Independent Learning	99	

Essential Material

- Abernethy, B. *_The Biophysical Foundations of Human Movement_*. 2nd ed.. Champaign, IL: Human Kinetics, 2005.

Supplementary Material

- Boone, T. *_Introduction to Exercise Physiology_*. Burlington, MA: Jones and Bartlett Learning, 2014.
- Gore, C.J. *_Physiological Tests for Elite Athletes_*. 2nd. Champaign, IL: Human Kinetics, 2013.
- Winter, E.M., A.M. Jones, R.R.C. Davison, P.D. Bromley and T.H. Mercer. *_Sport and Exercise Physiology Testing_*. UK: Routledge, 2007.

Requested Resources

- Room Type: Computer Lab